

Lentiviral Insert Generation

A complete step-by-step guide: from TCR genes to a finished annotated insert

This guide walks through the entire process of turning a TCR's V gene, J gene, and CDR3 into a finished lentiviral insert sequence. There are four stages: (1) set up the scripts, (2) run Stitchr to build each chain's nucleotide sequence, (3) trim the sequences and enter them into an input file, and (4) run the insert-generation script to produce the final output.

What You Need (one-time setup)

Before starting, make sure these are installed on your computer. You only do this once.

Tool	What it's for	Install command
Stitchr	Builds the full TCR nucleotide sequence from V/J/CDR3	<code>pip install stitchr</code> <code>stitchrdl -s HUMAN</code>
Insert scripts	Generates the lentiviral insert (included in this bundle)	No install – just download the folder
Python 3	Runs the scripts	Pre-installed on Mac (check: <code>python3 --version</code>)

Stage 1 — Download and Open the Scripts

Step 1.1 — Download the script folder

Download the insert-scripts bundle (the .zip that came with this guide) and unzip it. You'll get a folder containing these files:

- `lentiviral_insert_job.py` — *the main script you run*
- `functions.py`
- `generate_lentiviral_insert.py`
- `sequence_functions.py`

Put this folder somewhere easy to find, like your Desktop or Documents folder. All the other files you create (input and output) will go in this same folder.

Step 1.2 — Open Terminal and point it at the folder

Open the Terminal app. Then type `cd` followed by a space, and drag the script folder from Finder onto the Terminal window (this auto-fills the path). Press Enter. For example, if the folder is in Documents:

```
cd ~/Documents/GenerateInsertScripts
```

To confirm you're in the right place, type `ls` and press Enter. You should see the four script files listed.

Stage 2 — Run Stitchr for Each Chain

For each TCR you need two chains: alpha (TRA) and beta (TRB). Run Stitchr once per chain. The `-nl` flag is important — it tells Stitchr to skip the leader sequence, so the output starts right at the mature V region. This

also avoids the "leader not found" error.

Command format

```
stitchr -v [V gene] -j [J gene] -cdr3 [CDR3] -s HUMAN -l [TRA or TRB] -nl
```

Worked examples

Alpha chain (8.E3 TRA — TRAV2 / CAVDGSNTGKLIF / TRAJ37):

```
stitchr -v TRAV2 -j TRAJ37 -cdr3 CAVDGSNTGKLIF -s HUMAN -l TRA -nl
```

Beta chain (TRBV4-2 / CASSDRKTNEKLFF / TRBJ1-4):

```
stitchr -v TRBV4-2 -j TRBJ1-4 -cdr3 CASSDRKTNEKLFF -s HUMAN -l TRB -nl
```

Tip about gene names: drop the *01 allele suffix — Stitchr uses *01 by default, and the asterisk confuses the Terminal. Only keep a suffix if you specifically need a different allele (like *03), and then wrap the gene name in quotes:

```
stitchr -v "TRBV7-9*03" -j TRBJ2-7 -cdr3 CASSLPAYPQEYF -s HUMAN -l TRB -nl
```

Gene names with a slash (like TRAV29/DV5) also need quotes: -v "TRAV29/DV5"

When you run each command, Stitchr prints a FASTA sequence. Copy the nucleotide sequence (the long string of letters after the header line) — you'll trim it in the next stage.

Stage 3 — Trim the Sequences and Build the Input File

The Stitchr output contains the V/CDR3/J region followed by the constant region. The insert script needs only the V/CDR3/J part — it adds its own mouse constant regions. So you trim off the constant region from the 3' (right) end. The two chains trim differently:

Chain	Where to cut	Target length	Why
Alpha (TRA)	Cut right before the human TRAC region. The 'ATAT' sequence marks where it starts.	Remainder 0 (length divisible by 3)	The script needs remainder 0 for the alpha chain.
Beta (TRB)	Cut right before the human TRBC region, but KEEP the final G nucleotide.	Remainder 1 (ends in G)	The G forms the glutamate (E) that opens the constant region. Cutting it breaks the junction.

Step 3.1 — Create the input file

Make a plain-text file (in TextEdit, use Format → Make Plain Text; or use Excel and Save As → "Tab delimited Text"). It has a header row and one row per construct. The columns, separated by **Tabs**, are:

Column	What goes in it
sample	A short name for the construct (used in the output)
description	Optional notes (can be left blank)
cdr3_a_nt	The TRIMMED alpha nucleotide sequence (remainder 0)

cdr3_b_nt	The TRIMMED beta nucleotide sequence (remainder 1, ends in G)
cdr3_a	The alpha CDR3 amino acid sequence (for verification)
cdr3_b	The beta CDR3 amino acid sequence (for verification)

Important: the fields must be separated by real Tab characters, not spaces. Save the file with a `.txt` extension into the same folder as the scripts.

Step 3.2 — Move the file into the scripts folder

Make sure your finished input `.txt` file is saved in the same folder as `lentiviral_insert_job.py` and the other scripts.

Stage 4 — Generate the Insert

Step 4.1 — Run the script

In Terminal (still pointed at the scripts folder from Stage 1), run the script with **two** filenames: your input file, then a name for the output file.

```
python3 lentiviral_insert_job.py [input file] [output file]
```

For example:

```
python3 lentiviral_insert_job.py clone461_rank1.txt clone461_rank1_output.txt
```

Watch out: don't put spaces in your filenames. Use underscores instead (`clone461_rank1.txt`, not `clone461 rank1.txt`). A space makes the Terminal read it as two separate files. And remember you must supply both filenames — leaving off the output name gives an "index out of range" error.

Step 4.2 — Check the output

The script writes the output file into the same folder. Open it to see the full insert nucleotide sequence and its translated amino acid sequence. To confirm it worked, check that both CDR3 sequences appear in the amino acid sequence, and that the beta junction reads correctly (the sequence around the beta/constant boundary should contain EDLRNV with an E right at the seam, not a stray K or Q).

Note: you may see a short string of extra letters after a stop symbol at the very end of the amino acid sequence. This is the script over-translating the scaffold past the stop codon and is harmless — it is not part of the real protein.

Optional — Making the Output Look Nice

The raw output file is plain text. To turn it into a color-coded annotated document (with the beta chain, alpha chain, restriction sites, and scaffold regions highlighted), the sequence can be formatted into a Word/PDF document. This is currently done as a separate formatting step. The annotated version makes it much easier to read the construct and spot the cloning sites and junctions at a glance.

Quick Reference — The Whole Process

Stage	Command / Action
1. Open folder in Terminal	<code>cd ~/Documents/GenerateInsertScripts</code>
2. Run Stitchr (alpha)	<code>stitchr -v TRAV2 -j TRAJ37 -cdr3 [CDR3] -s HUMAN -l TRA -nl</code>
Run Stitchr (beta)	<code>stitchr -v TRBV4-2 -j TRBJ1-4 -cdr3 [CDR3] -s HUMAN -l TRB -nl</code>
3. Trim + build input.txt	Alpha -> remainder 0 (cut at ATAT) Beta -> remainder 1 (keep final G)
4. Run insert script	<code>python3 lentiviral_insert_job.py input.txt output.txt</code>

Common Errors

Error message	Cause & fix
no matches found: TRBV7-9*01	The * confused the Terminal. Drop the *01, or wrap the gene name in quotes.
LEADER sequence not found	You forgot the -nl flag. Add it to the Stitchr command.
list index out of range	You forgot the output filename. Supply both: input.txt AND output.txt.
No such file or directory	Filename has a space, or the file isn't in this folder. Use underscores; check with ls.
Mismatch! (in output)	A row in your input file has the wrong number of columns. Check your Tabs.